IA9100 DRAFT 7.1.5.1 NOTE · MEASUREMENT · NEW

The gauge nobody studied

New in the draft: measurement variation is understood through gauge R&R or attribute studies.

1. List the key characteristics on your three highest-volume parts
2. Name the gauge that measures each one
3. Find the study on file for each gauge
4. Judge each result against your criteria
5. Run one study this week
6. Act on a study that came back unacceptable

Swipe: one action per page, with the record that proves it.

ACTION 1 OF 6 · DRAFT 7.1.5.1 note

NEW IN THE DRAFT

List the key characteristics on your three highest-volume parts

Start where the volume is. Take them from the drawing and the control plan, not from memory.

THE RECORD THAT PROVES IT

- ☐ The control plan for each part, key characteristics marked

> **Sample plant:** Its five control plans each list bore diameter and hardness among the characteristics.

ACTION 2 OF 6 · DRAFT 7.1.5.1 note

NEW IN THE DRAFT

Name the gauge that measures each one

In-process or at final inspection. Use the gauge ID from your calibration register, so the study, the gauge and its calibration can be matched.

THE RECORD THAT PROVES IT

- ☐ A characteristic-to-gauge list, by gauge ID
- ☐ The calibration register entry for each gauge

Watch for: A gauge that measures a key characteristic but isn't on the calibration register.

NEW IN THE DRAFT

Find the study on file for each gauge

A gauge R&R for gauges that read a value, an attribute study for go/no-go checks. Note the date and the result.

THE RECORD THAT PROVES IT

- ☐ The gauge R&R or attribute study, dated
- ☐ The result, and who accepted it

> **Sample plant:** Three gauge studies on file: bore gauge BG-6601, hardness tester HT-2205 and CMM-4401.

ACTION 4 OF 6 · DRAFT 7.1.5.1 note

NEW IN THE DRAFT

Judge each result against your criteria

Your own acceptance criteria and your customer's. A study filed without a decision isn't finished.

THE RECORD THAT PROVES IT

- ☐ The study with its accept or reject decision
- ☐ The criteria it was judged against

Watch for: A study nobody signed off. It shows the work, not the decision.

NEW IN THE DRAFT

Run one study this week

Pick the gauge with no study that measures the tightest tolerance, and study it with the people who use it.

THE RECORD THAT PROVES IT

- ☐ The new study, dated, with the parts and appraisers used

> **Why that gauge:** On the tightest tolerance, gauge variation uses up the most of the band.

ACTION 6 OF 6 · DRAFT 7.1.5.1 note

NEW IN THE DRAFT

Act on a study that came back unacceptable

Fix the gauge or the method before you trust the process numbers, then study it again.

THE RECORD THAT PROVES IT

- ☐ The action, with an owner and a date
- ☐ The repeat study

> **Watch for:** A study that came back unacceptable and nothing changed afterwards: the record shows you knew.

THIS WEEK

Pick one. Keep the record.

1. **Find the gauges with no study**
   Characteristic-to-gauge list
2. **Study the tightest-tolerance one**
   The new gauge R&R, dated
3. **Close out any unacceptable result**
   Action and repeat study

TELL ME IN THE COMMENTS

Has a gauge R&R ever changed how you inspect a part?

IA9100 is not yet final. Clauses are from the IAQG's published key-change summaries (November 2023), as the draft numbers them.